

Statistics in Health Sciences - Assignment 1

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28th October 2026

1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al. to the data set `copd` in the file `copd redux.RData`. Data `copd` is a subset of 45 variables of the original data analyzed in the cited paper.

2 Analysis

2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

“The subset used here included 342 participants and 85 variables. The range of missing data in the different variables went from 0% to 47.7%, with 68.2% of the variables having less than 5% of the data missing. Only 13.7% of the participants had complete data on all 85 variables. Overall, 5.9% of the information was missing; that is, 5.9% of the total of 29,070 cells ($85 \times 342 = 29,070$) had missing data.”

Instructions:

- Note that, since you are working with a subset of the original data set, red numbers above are not expected to be the same than in your results.
- In your `Rnw` document, the text above must be able to automatically update all red numbers in the case of changing the values of `copd`. I.e., if data change, then all red numbers must also automatically update accordingly in your pdf when compiling your `Rnw`. As a hint, you can follow this procedure:
 1. For each red number, create an R object, which will depend on data and gives the desired quantity (i.e., if data change, the value in the object changes accordingly),
 2. To force rounding a quantity to a given decimal precision, you can use `sprintf()` (as seen in Exercise 0). Do not use `round()`,
 3. To get the thousands comma separator as in “29,070”, you can use `prettyNum()`,
 4. Once you have an R object for each red number formatted as desired, use `\Sexpr{}` to insert each of them inline in $\text{\LaTeX}$ (as seen in Exercise 0).

```
> n_obs <- nrow(copd) # Number of participants
> n_vars <- ncol(copd) # Number of variables
> nans <- 100*colMeans(is.na(copd)) # Percentage of missing data by variable
> min_nan <- sprintf("%.2g", min(nans)) # MIRAR # Lower percentage
> max_nan <- sprintf("%.2g", max(nans)) # Upper percentage
>
> # Variables with less than 5% of data missing (percentage)
> less_5 <- sprintf("%.2g", 100*length(nans[nans<5])/n_vars)
>
> # Percentage of complete cases
> comp_cases <- sprintf("%.2g", 100*mean(complete.cases(copd)))
>
> # Percentage of missing data
> miss_total <- sprintf("%.2g", 100*mean(is.na(copd)))
>
> # Total cells observed (variables*observations)
> total_cells = prettyNum(n_obs*n_vars, big.mark = ",", decimal.mark = ".")
```

“The subset used here included 342 participants and 45 variables. The range of missing data in the different variables went from 0% to 32%, with 69% of the variables having less than 5% of the data missing. Only 33% of the participants had complete data on all 45 variables. Overall, 5.4% of the information was missing; that is, 5.4% of the total of 15,390 cells ($45 \times 342 = 15,390$) had missing data.”

2.2 Multiple imputation

Apply multiple imputation to copd data using 100 imputations. Explain in a reasoned manner how you decided to set the values of the arguments `method` and `predictorMatrix` when using the `mice()` function. Show your code.

```
> # Display of default imputation methods for each variable
> ini <- mice(copd, maxit=0)
> ini

## Class: mids
## Number of multiple imputations: 5
## Imputation methods:
##          id index_barthel2v1 quo_calcpstv1 rv_prepv1 incp_fevv1
##          ""                  "pmm"          ""          "pmm"          "pmm"
## fevp_postv1 rv_tlc1v1 ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1
##          ""                  "pmm"          "pmm"          "pmm"          "pmm"
## sin11_v1 vas2_v1 ic_prepv1 iam_chv1 sin15_v1
## "logreg"          "pmm"          "pmm"          "logreg"          "logreg"
## cil8_s_mv1 sin7_v1 sin8_v1 ch_v1 vc_prepv1
## "pmm"          "logreg"          "logreg"          "pmm"          "pmm"
## dtdve_ecov1 cccs1_v0 kcals_set_v0 impactsv1 monoA_v1
## "pmm"          "logreg"          "pmm"          "pmm"          "pmm"
## lvmass_ase_ecov1 vtit_catv1 urg_ambv0 po2_v1 ahpv_v1
## "pmm"          "logreg"          ""          "pmm"          "pmm"
## pro_v1 basA_v1 fc_pm1 tad_v1 col_v1
## "pmm"          "pmm"          "pmm"          "pmm"          "pmm"
## sin5_v1 bc_v1 fvc1_postv1 vas1_v1 sin6_v1
## "logreg"          "pmm"          ""          "pmm"          "logreg"
## graudisn_v1 cpcr_mg_l_med1v1 neutA_v1 ahpl_v1 activityv1
## "polyreg"          "pmm"          "pmm"          "pmm"          "pmm"

## PredictorMatrix:
##          id index_barthel2v1 quo_calcpstv1 rv_prepv1 incp_fevv1 fevp_postv1
## id          0          1          1          1          1          1
## index_barthel2v1 1          0          1          1          1          1
## quo_calcpstv1    1          1          0          1          1          1
## rv_prepv1        1          1          1          0          1          1
## incp_fevv1       1          1          1          1          0          1
## fevp_postv1      1          1          1          1          1          0
##          rv_tlc1v1 ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_prepv1
## id          1          1          1          1          1          1          1
## index_barthel2v1 1          1          1          1          1          1          1
## quo_calcpstv1    1          1          1          1          1          1          1
## rv_prepv1        1          1          1          1          1          1          1
## incp_fevv1       1          1          1          1          1          1          1
## fevp_postv1      1          1          1          1          1          1          1
##          iam_chv1 sin15_v1 cil8_s_mv1 sin7_v1 sin8_v1 ch_v1 vc_prepv1 dtdve_ecov1
## id          1          1          1          1          1          1          1          1
## index_barthel2v1 1          1          1          1          1          1          1          1
## quo_calcpstv1    1          1          1          1          1          1          1          1
## rv_prepv1        1          1          1          1          1          1          1          1
## incp_fevv1       1          1          1          1          1          1          1          1
## fevp_postv1      1          1          1          1          1          1          1          1
##          cccs1_v0 kcals_set_v0 impactsv1 monoA_v1 lvmass_ase_ecov1 vtit_catv1
## id          1          1          1          1          1          1          1          1
## index_barthel2v1 1          1          1          1          1          1          1          1
## quo_calcpstv1    1          1          1          1          1          1          1          1
## rv_prepv1        1          1          1          1          1          1          1          1
## incp_fevv1       1          1          1          1          1          1          1          1
## fevp_postv1      1          1          1          1          1          1          1          1
##          urg_ambv0 po2_v1 ahpv_v1 pro_v1 basA_v1 fc_pm1 tad_v1 col_v1 sin5_v1
## id          1          1          1          1          1          1          1          1          1
## index_barthel2v1 1          1          1          1          1          1          1          1          1
## quo_calcpstv1    1          1          1          1          1          1          1          1          1
## rv_prepv1        1          1          1          1          1          1          1          1          1
## incp_fevv1       1          1          1          1          1          1          1          1          1
## fevp_postv1      1          1          1          1          1          1          1          1          1
##          bc_v1 fvc1_postv1 vas1_v1 sin6_v1 graudisn_v1 cpcr_mg_l_med1v1 neutA_v1
## id          1          1          1          1          1          1          1          1          1
## index_barthel2v1 1          1          1          1          1          1          1          1          1
```

```
## quo_calcpstv1      1      1      1      1      1      1      1
## rv_prepv1          1      1      1      1      1      1      1
## incp_fevv1         1      1      1      1      1      1      1
## fevp_postv1        1      1      1      1      1      1      1
##                   ahpl_v1 activityv1
## id                  1      1
## index_barthel2v1    1      1
## quo_calcpstv1       1      1
## rv_prepv1           1      1
## incp_fevv1          1      1
## fevp_postv1         1      1
```

We can see that the mice package suggest imputing initially with de mean for continuous variables and with a logistic regression for binary ones. Further research on the dataset has shown that the variables have no deterministic relation. So we do not change the imputation methods. We have a variable (id) that don't have any role in the imputation process, so we change `predictorMatrix` and `methods` according to that.

```
> # Change in parameters
> meth <- ini$method
> pred <- ini$predictorMatrix
>
> meth["id"] <- "" # do not impute "id"
> pred["id", ] <- 0 # we don't want "id" to be imputed
> pred[, "id"] <- 0 # we don't want "id" to be predictor
```

```
> # Imputation of 100 datasets
> M <- 2 # Number of imputations for each missing value
> imp <- mice(data = copd,
+             m = M,
+             method = meth,
+             predictorMatrix = pred,
+             maxit = 20,
+             printFlag = TRUE,
+             seed = 666)
```

```
##
## iter imp variable
## 1 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 1 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 2 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 2 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 3 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 3 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 4 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 4 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 5 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 5 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 6 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 6 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 7 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 7 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 8 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 8 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 9 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 9 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 10 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 10 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 11 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 11 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 12 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 12 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 13 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 13 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 14 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 14 2 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
## 15 1 index_barthel2v1 rv_prepv1 incp_fevv1 rv_tlcvt ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_p
```

##	15	2	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	16	1	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	16	2	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	17	1	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	17	2	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	18	1	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	18	2	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	19	1	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	19	2	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	20	1	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic
##	20	2	index_barthel2v1	rv_prepv1	incp_fevv1	rv_tlc1v1	ctnf_s_mv1	vtdvdbid_ecov1	pp_ecov1	sin11_v1	vas2_v1	ic